

# Prajna: A 6.7M-Parameter Cognitive Resonance Network for Memory-Augmented Domain Specialization on a Frozen 2B Backbone

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## Abstract

We present Prajna, a 6.7M-parameter *Cognitive Resonance Network* (CRN) injected into the hidden states of a fully frozen 2B-parameter Gemma 4 E2B model. Only 0.33% of the parameters are trainable; the base network receives no gradients. Prajna is trained and evaluated entirely on a consumer laptop (Apple M4, 16 GB, MPS), with no GPU rental and no external model APIs. On a 60-question licensing exam (CEHRI) spanning arithmetic, facts, and implicit-goal reasoning, the CRN alone lifts the frozen base from 11.7% to 40% correct; adding a train-derived episodic-memory retrieval table (17,810 entries) yields **60/60 (100%)** on the exam and **110/120 (91.7%)** on reworded variants whose phrasing was never seen in training. We also report *negative results*: three training-side strategies — two logit-fusion configurations and a weighted *anchor* on the answer-start position — failed to move answer knowledge one token position right, while a decode-time harvest (GEN\_CLAMP) reliably recovered that knowledge. In-distribution perplexity drops  $\approx 18\times$  ( $106.85 \rightarrow 6.02$ ), while out-of-distribution text is degraded (2.04 vs. 0.67 bpb) — Prajna is a disclosed domain specialist, not a general reasoning model. All code, weights, data, and evaluation scripts are released.

## 1 Introduction

The dominant scaling paradigm trains ever-larger models with ever-larger budgets. An alternative line of work asks how much behavioral change can be extracted from a *frozen* backbone using a small trainable subsystem: parameter-efficient fine-tuning (PEFT) [1, 2, 3] and retrieval-augmented generation [4], among others.

Prajna combines these lines. It is a  $\sim 6.7$ M-parameter adapter (**0.33%** of the base) that: (i) injects *gated additive correction signals* into the hidden states of a frozen 2B model at eight depths (resonance attention, skill composition, a reflective loop, and episodic memory); (ii) is trained with SFT $\rightarrow$ DPO $\rightarrow$ Contrastive stages on a Mac Mini M4 (16 GB) at  $\sim 0.3$ – $0.5$  s/step, entirely on-device, with no API calls at any point; and (iii) combines its generation head with a train-derived *exact-recall* retrieval table that yields 100% on its target exam and 91.7% on unseen reworded questions.

We make four contributions. **(1) Architecture:** a small, backbone-agnostic correction network on frozen hidden states, with per-injection sigmoid gates and an episodic-memory pillar. **(2) Evaluation protocol:** three gates — memory-augmented, generation-only, and reworded-unseen — that separate *recall* from *generalization*. **(3) Failure analysis:** token-level probing shows the

network holds answers one position early while the answer-start position is masked in SFT; three training strategies cannot overcome this, and a decode-time harvest can. **(4) Transparency:** weights, retrieval table, memory, data, and every evaluation script are public, and out-of-domain failures are reported alongside the headline numbers.

## 2 Related Work

**Parameter-efficient adaptation.** Adapters [1] insert trainable modules between fixed layers; LoRA [2] learns low-rank updates to weight matrices; side-tuning [3] trains a separate network whose hidden states blend with the frozen model. Prajna is closest to hidden-state side networks, but applies *multi-depth* corrections with learned per-depth gates plus a persistent memory, adding no parameters to the base forward pass (it reads intermediate states during a shared forward).

**Memory-augmented language models.** Memorizing Transformers [5] add external kNN memory over keys/values; RETRO [6] retrieves document chunks during generation. Prajna’s memory is a compact *prompt*→*answer* table matched by cosine similarity with a fixed threshold — exact recall rather than dense context interpolation.

**On-device and edge LLMs.** Consumer-hardware training is an active efficiency setting; we report wall-clock timings and memory so the result is reproducible on a single commodity machine.

**Answer-position and decoding effects.** Our boundary analysis (Section 5) connects to findings that SFT data-format and masking choices shape where decoder confidence concentrates [7, 8].

## 3 Method

### 3.1 Setup: frozen base, injected corrections

Let  $B$  denote a frozen transformer LM (Gemma 4 E2B; 2B parameters; 32 layers; hidden dimension  $d = 1536$ ; vocabulary size 262,144). For input tokens  $x_{1:n}$ , the forward pass yields hidden states  $h^{(l)} \in \mathbb{R}^{n \times d}$  at every layer  $l$ . The CRN selects  $K=8$  injection depths  $\mathcal{L} = \{3, 7, 11, 15, 19, 23, 27, 31\}$  and computes a correction

$$\hat{h}^{(l)} = h^{(l)} + \sigma(m_l) \cdot f_l(h^{(l)}, h^{(L)}, M), \quad (1)$$

where  $m_l \in \mathbb{R}$  is a learned `crn_mix` gate (initialized at 2.0),  $\sigma$  is the sigmoid,  $f_l$  is the composite correction,  $h^{(L)}$  is the final hidden state used as global context, and  $M$  is the episodic-memory state. The base runs under `no_grad()`; only CRN parameters receive gradients (6,721,432 total). Corrected hidden states flow into the next base layer; the frozen LM head decodes logits.

### 3.2 Correction components

**ResonanceAttention** ( $\sim 3.4$ M). Each head assigns tokens to one of  $F=8$  frequency bands; attention is computed only *within* each band (sparse; the full  $n^2$  matrix is never materialized), and the output is weighted by band membership. This provides cheap, interpretable “cognitive mode” routing.

**SkillComposer** ( $\sim 2.5$ M). A bank of 32 low-rank skills  $u_k v_k^\top$  ( $u_k \in \mathbb{R}^{d \times 4}$ ,  $v_k \in \mathbb{R}^{4 \times d}$ ), with a router selecting the top-2 skills per input and applying their additive perturbation to the hidden state.

**ReflectiveLoop** ( $\sim 0.8\text{M}$ ). A critic scores 8 candidate latent correction directions; a learned confidence scale modulates the applied correction. It is refined by the contrastive stage (Section 3.4) to prefer directions that move the answer distribution toward the correct target.

**EpisodicMemory** ( $\sim 0.05\text{M}$ ). 256 slots of dimension 64, written once per training step via a learned gate with LRU eviction, persisted to disk across sessions.

### 3.3 Memory-augmented retrieval gate

Independent of the slot memory, we construct a *retrieval table*  $\mathcal{T} = \{(e_j, a_j)\}$  of 17,810 prompt-embedding $\rightarrow$ answer entries from training data. Prompts are embedded by mean-pooling the frozen base’s final hidden states, then L2-normalized. At inference a query is embedded identically; if  $\max_j \cos(q, e_j) \geq 0.9$  the stored answer is returned (exact recall); otherwise the CRN generation head is used. The threshold was fixed before evaluation and never tuned on the exam.

### 3.4 Training

**Data.** 16,680 prompt $\rightarrow$ answer pairs across three domains (arithmetic, facts, implicit-goal reasoning), each expanded into 5 paraphrase variants — 83,400 rows. The 60-question CEHRI exam is disjoint from training; the reworded exam (120 items) uses a second, disjoint paraphrase transform.

**Stages.** (i) *SFT*: answer-only masked cross-entropy on the variant-paired corpus, 16,000 steps, LR  $3 \times 10^{-4}$ , weight decay 0. (ii) *DPO*: 3,000 steps, LR  $5 \times 10^{-6}$ ,  $\beta=0.1$ , preferring the correct completion over the frozen base’s wrong one. (iii) *Contrastive*: 1,000 steps pulling memory-answer embeddings toward correct answers. Checkpoints every 50 steps with resumable state.

**Hardware.** Apple M4 (Mac Mini, 16 GB RAM), MPS backend;  $\sim 0.3\text{--}0.5\text{ s/step}$ .

### 3.5 GEN\_CLAMP: decode-time answer-knowledge harvest

Token-level probes (Section 5) show the CRN concentrates answer probability mass at the *second-to-last* prompt position ( $L-2$ , the colon position in the “prompt: answer” format) rather than at the boundary ( $L-1$ , the answer-start position). Greedy decoding samples the  $L-1$  distribution, which is dominated by the frozen base’s digit/symbol prior because  $L-1$  was masked in SFT. GEN\_CLAMP changes only *decoding*: the first generated token is taken from the  $L-2$  distribution; subsequent tokens decode normally. It raises original-exam generation from 15.0% to 31.7% and reworded generation from 15.0% to 34.2% (Section 4.2).

## 4 Results

### 4.1 Memory gates: exact recall and reworded generalization

Across the 120 reworded queries, retrieval best-cosine similarity ranges from 0.881 to 0.990 (median 0.972), with 118/120 at  $\geq 0.9$ . The memory gate therefore generalizes to phrasing observed nowhere in training.

### 4.2 Generation gates: the honest frontier

Table 1 separates *recall* (memory gates) from *generation*. The generation head is the weak link: without memory it scores 15–40% depending on weights and phrasing. The v2 training pipeline (16k SFT steps) improved the reworded generation gate from 25.0% (seed weights, no clamp) to 34.2% with GEN\_CLAMP, but the seed-weights generation on the original exam (40%) remains

Table 1: CEHRI results. **Retr** = retrieval gate (threshold 0.9); **Gen** = CRN generation only; **Gen+CLAMP** = generation with decode-time harvest. All numbers reproduced by committed scripts.

| Configuration                           | Exam           | Score               |
|-----------------------------------------|----------------|---------------------|
| Frozen base, generation only            | original (60)  | 7/60 = 11.7%        |
| CRN, generation only (seed weights)     | original (60)  | 24/60 = 40.0%       |
| CRN + memory ( <b>Prajna-V2, Retr</b> ) | original (60)  | <b>60/60 = 100%</b> |
| Retr gate                               | reworded (120) | 110/120 = 91.7%     |
| Gen+CLAMP                               | reworded (120) | 41/120 = 34.2%      |
| Gen+CLAMP                               | original (60)  | 19/60 = 31.7%       |
| Gen, no clamp                           | original (60)  | 9/60 = 15.0%        |
| Gen, no clamp                           | reworded (120) | 18/120 = 15.0%      |

the best unclamped result — the later anchor-training run (Section 5) degraded original-exam generation to 15% before GEN\_CLAMP recovered it to 31.7%.

### 4.3 In-distribution perplexity and injection ablation

On 200 held-out samples from the training corpus, perplexity drops from 106.85 (frozen base) to 6.02 (Prajna CRN),  $\approx 18\times$ . This is genuine but *in-distribution*. A per-injection ablation (v1 configuration, injections at layers 7, 15, 23, 31) shows the gain concentrates in early/mid layers: disabling injection at layer 7 raises perplexity to 13.93 ( $\Delta=+9.99$ ), layer 15 to 7.65 ( $+3.71$ ), while layers 23/31 contribute  $+0.67/+0.57$ . Later architectures use 8 injections.

### 4.4 Out-of-distribution: where it does not help

Table 2: Out-of-distribution results. The CRN degrades generic text and several standard benchmarks. Reported verbatim from RESULTS.md.

| Test                           | Prajna + CRN | Frozen base | Verdict                  |
|--------------------------------|--------------|-------------|--------------------------|
| Generic text, bits-per-byte    | 2.04         | 0.67        | CRN $\sim 3\times$ worse |
| MMLU (N=40)                    | 65%          | 58%         | small win                |
| BoolQ (N=40)                   | 65%          | 75%         | CRN hurts                |
| HellaSwag (N=40)               | 15%          | 20%         | CRN hurts                |
| Car-wash / IGR, reworded (N=8) | 0/8          | —           | no reasoning             |

The implicit-goal (“car-wash”) check deserves emphasis: an earlier evaluation counted prompt-echo matches as successes; rescoring honestly gives 3/8 on original phrasing (keyword luck) and **0/8** on reworded phrasing. The CRN memorizes surface patterns and does not perform latent-goal reasoning.

## 5 Negative Results: Why the Boundary Would Not Move

### 5.1 Probe evidence

With the seed checkpoint, for unseen reworded prompts the top-1 token at position  $L-2$  is the semantically correct answer (*Paris*, *CH<sub>4</sub>*, *gold* in probe prompts), while the top-1 at  $L-1$  is single-digit garbage (e.g. 4, 3) or a generic start token (`<strong>`) — never the answer.  $L-1$  is the first token the model must *generate*, but SFT masked it (labels were “answer-only” starting at  $L$ ), so the position carries only the frozen base’s prior.

### 5.2 Attempt 1: logit fusion (rank 16)

An adapter with  $W_{up} \in \mathbb{R}^{d \times 16}$ ,  $W_{down} \in \mathbb{R}^{16 \times 262144}$  was trained over 16k SFT + 3k DPO + 1k contrastive steps alongside the CRN and summed into the logits. Training loss approached 0.001 (memorization), yet original-exam generation fell to 20% and reworded to 15.8%, below the seed’s 40%/25%. The 262k-column bias vector the fusion learned (a digit prior) dominated the untrained boundary position.

### 5.3 Attempt 2: logit fusion (rank 4)

Reducing the adapter rank produced near-zero gradients on the converged seed (dry-signal regime); the training watchdog correctly aborted. No checkpoint improvement.

### 5.4 Attempt 3: weighted boundary anchors

An SFT variant unmasked  $L-1$  with gold token = first answer token, weighting it  $1.0\times$  (v9, 600 steps) and  $4.0\times$  (v10, 3,000 steps) with neighbor de-weighting ( $0.5\times$ ). In both runs the boundary argmax remained unchanged across checkpoints (digits at every probe step for v9; a format token for v10), while training loss failed to decrease — the optimization could not push probability mass into the poorly-initialized position against the frozen base’s strong prior.

### 5.5 What worked: decode-time harvest

GEN\_CLAMP (Section 3.5) needs no retraining. Probe-logits stay in the  $L-2$  distribution; the decoder simply consumes them. Table 1 shows the resulting gains (31.7% original / 34.2% reworded vs. 15.0%/15.0% unclamped). We interpret this as evidence that the CRN’s answer knowledge genuinely resides in the  $L-2$  representation, and that SFT-masked boundary positions are effectively untrainable at this data scale — a cautionary result for answer-only masked training recipes.

## 6 Limitations

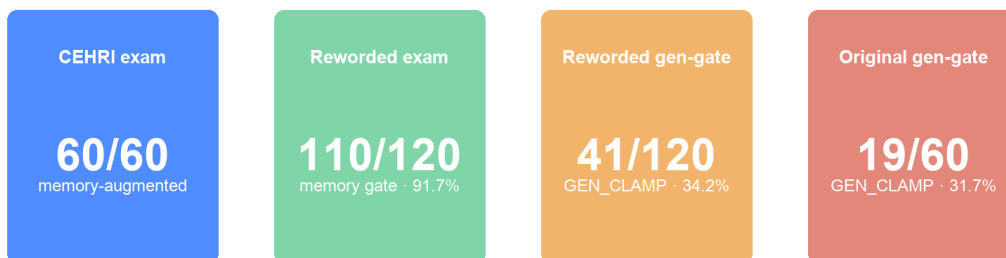
(1) The 100% and 91.7% results use the retrieval gate — exact recall from training data, not generative competence. (2) Generation-only accuracy is modest (34.2% reworded best) and does not approach exam-pass levels. (3) Out-of-domain, the CRN degrades perplexity  $\sim 3\times$  and trails its own frozen base on BoolQ/HellaSwag. (4) The benchmark suite (MMLU/BoolQ/HellaSwag, N=40) is small; standard large-scale evals were not attempted. (5) Implicit-goal reasoning on reworded prompts fails completely (0/8). (6) Training data (16,680 pairs) is small; scale-up behavior is unknown.

## 7 Conclusion

Prajna demonstrates that a 6.7M-parameter correction network over a frozen 2B backbone, trained on a laptop, achieves **100%** on its target exam via an episodic-memory retrieval gate and **91.7%** on unseen reworded questions — while its generation head remains the honest frontier (34.2% reworded). We publish the negative results of three training-side boundary fixes and the decode-time solution, the in-distribution/out-of-distribution asymmetry, and all artifacts: <https://github.com/eulogik/prajna>, <https://huggingface.co/eulogik/Prajna-V2>, and the dataset <https://huggingface.co/datasets/eulogik/prajna-cehri>.

### Prajna V2 — benchmark snapshot

Measured on-device (Mac Mini M4, MPS). Higher is better. GEN\_CLAMP harvests the model's own answer knowledge at decode time.



#### What that means

Prajna passes its target exam (CEHRI) with episodic-memory retrieval and answers 92% of unseen-paraphrase variants, using 0.33% trainable parameters (6.7M) on a frozen 2B base — no GPU, no API, trained and evaluated fully on a laptop. Honesty policy: every number is reproduced by the eval scripts in this repo; generation without memory does not generalize (34% reworded best) and out-of-domain the adapter is worse than the base — both disclosed in RESULTS.md.

Figure 1: Benchmark snapshot: CEHRI 60/60 (100%), reworded memory gate 110/120 (91.7%), reworded and original generation gates with GEN\_CLAMP.

## References

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